

Operating Systems: Process Scheduling

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1. Introduction

An operating system is system software that manages computer hardware and provides services for application programs. A process is a program in execution. The operating system keeps track of processes and decides when each process should receive processor time.

A scheduler selects a process from a ready queue and determines which process will execute next. The choice of scheduling policy affects responsiveness, waiting time, turnaround time, and processor utilization.

2. CPU Scheduling

CPU scheduling is the mechanism used by an operating system to select a process for execution when multiple processes are ready to run. A scheduling algorithm defines the rule used to make this selection.

3. First-Come, First-Served (FCFS)

First-Come, First-Served scheduling assigns the CPU to processes in the order in which they enter the ready queue. FCFS is simple and non-preemptive. However, a long process at the front of the queue can make shorter processes wait for a long time.

4. Shortest Job First (SJF)

Shortest Job First selects the ready process with the smallest expected CPU burst. SJF can minimize average waiting time when burst lengths are known or estimated accurately. The algorithm may cause starvation when long processes continue to wait while shorter processes are repeatedly selected.

5. Shortest Remaining Time First (SRTF)

Shortest Remaining Time First is the preemptive form of Shortest Job First. When a new process arrives with a shorter remaining CPU burst than the currently running process, the scheduler can preempt the current process and execute the new one.

6. Priority Scheduling

Priority scheduling assigns each process a priority value. The scheduler selects a process according to priority. Depending on the implementation, priority scheduling may be preemptive or non-preemptive. A common problem is starvation: a low-priority process may wait indefinitely if higher-priority processes continually arrive.

Aging is a technique used to reduce starvation. The priority of a waiting process is gradually increased so that it eventually receives processor time.

7. Round Robin Scheduling

Round Robin scheduling is designed for time-sharing systems. Each ready process receives a fixed time quantum. When the quantum expires, the running process is preempted and placed at the end of the ready queue if it has not finished.

A very small time quantum can cause frequent context switches. A very large time quantum makes Round Robin behave more like First-Come, First-Served.

8. Scheduling Criteria

Criterion	Meaning
Waiting time	Time a process spends waiting in the ready queue.
Turnaround time	Time from process submission to process completion.
Response time	Time from a request until the system first produces a response.
Throughput	Number of processes completed per unit of time.
CPU utilization	Percentage of time the processor is kept busy.

9. Worked Example

Consider three processes named P1, P2, and P3. Their CPU burst times are 5 ms, 2 ms, and 8 ms respectively. The values are used only to illustrate how scheduling algorithms can produce different execution orders.

For FCFS, the processes are considered in arrival order. For SJF, the process with the shortest burst is considered first. For Round Robin, execution is divided according to the selected time quantum.

10. Common Discussion Terms

In classroom discussions, students may encounter words such as example, question, answer, result, method, problem, chapter, section, table, and page. These words help organize explanations but are not necessarily operating-system concepts. A knowledge graph should distinguish such incidental words from technical concepts that the material actually explains.

11. Summary

CPU scheduling determines which ready process receives processor time. FCFS is simple and non-preemptive. SJF selects the shortest expected CPU burst. SRTF adds preemption to SJF. Priority scheduling selects processes according to priority and may require aging to reduce starvation. Round Robin uses a time quantum and is commonly associated with time-sharing systems. Scheduling policies can be compared using waiting time, turnaround time, response time, throughput, and CPU utilization.